Symmetry Method Homework 2

1) $\mathbb{Z}_3$ subgroup of S_3

Consider $S_3 = \{\mathbb{1}, R_1, R_2, F_1, F_2, F_3\}$ and the subgroup
$\mathbb{Z}_3 = \{\mathbb{1}, R_1, R_2\}$

1.1) Find all right cosets & left cosets to show that
$\mathbb{Z}_3$ is a normal (invariant) subgroup of S_3

1.2) Find all conjugates elements of S_3
and group them into conjugacy classes

2). Quarternion

The Quaternion group Q_8 has 8 elements
$\{1, i, j, k, -1, -i, -j, -k\}$ with the usual multiplication.

2.1) Find all conjugacy classes

2.2) Find the normal subgroup of order 2
(subgroup with 2 elements)

2.3) Find all normal subgroups of order 4
(Subgroups with 4 elements)